temperature was above 10 billion billion billion (10^{28}) degrees—some thousand billion billion times the temperature at the center of the sun—extreme conditions that existed prior to 10^{-35} seconds after the bang. Above that temperature, these physicists suggested, photons, gluons of the strong force, as well as W and Z particles, could all be freely interchanged with one another—a more robust gauge symmetry than that of the electroweak theory—without any observable consequence. Georgi and Glashow thus suggested that at these high energies and temperatures there was complete symmetry among the three nongravitational-force particles, and hence complete symmetry among the three nongravitational forces.[11]

Glashow and Georgi's grand unified theory went on to say that we do not see this symmetry in the world around us—the strong nuclear force that keeps protons and neutrons tightly glued together in atoms seems completely separate from the weak and electromagnetic forces—because as the temperature dropped below 10^{28} degrees, another species of Higgs field entered the story. This Higgs field is called the *grand unified Higgs.* (Whenever they might be confused, the Higgs field involved in electroweak unification is called the *electroweak Higgs.*) Similar to its electroweak cousin, the grand unified Higgs fluctuated wildly above 10^{28} degrees, but calculations suggested that it condensed into a nonzero value when the universe dropped below this temperature. And, as with the electroweak Higgs, when this grand unified Higgs ocean formed, the universe went through a phase transition with an accompanying reduction in symmetry. In this case, because the grand unified Higgs ocean has a different effect on gluons than it does on the other force particles, the strong force splintered off from the electroweak force, yielding two distinct nongravitational forces where previously there was one. A fraction of a second and a drop of billions and billions of degrees later, the electroweak Higgs condensed, causing the weak and electromagnetic forces to split apart as well.

While a beautiful idea, grand unification (unlike electroweak unification) has not been confirmed experimentally. To the contrary, Georgi's and Glashow's original proposal predicted a trace, residual implication of the universe's early symmetry that should be apparent today, one that would allow protons to every so often transmute into